TieredKV: Two-Tier KV Cache

Phase 0: Initialization (Once at Prefill)

Prompt Tokens (prefill)

Initial Bifurcation — Observation-Window Scores
Actual-query option; key proxy by default

Tier 1: GPU VRAM	Tier 2: Cold KV	Dropped Tokens
Hot working set	Simulated STT-RAM tier	Not a storage tier
Sinks + recent + top scores	Live rows + later shadows	Beyond both budgets
Configurable hot budget	Configurable cold budget	Permanently discarded

Per Decode Step (after the first prompt prediction)

1. Prepare Visible KV

- Hot rows + quota-limited live cold rows; shadows excluded
- Recency selects cold exposure; same count across layers
- Native option: gather selected cold KV from pinned CPU memory
- Use this prepared set in the next model forward

2. Model Attention & Feedback

- Attend hot + selected live cold + current token
- Produce next-token logits and real attention weights
- Update hot scores and cold access timestamps

3. Promote Attention-Ranked Cold Rows

- Apply promotion budget, re-entry gate and live-cold floor
- Inclusive mode: keep a cold shadow backup
- Exclusive ablation: remove the promoted cold copy

4. Evict & Demote

a) Admit current-token KV; protect sinks and recent tokens

b) Hot overflow: demote the lowest age-normalized score
Optional write-aware weight favors rows with backups

Shadow backup?
- YES → Reactivate backup; no KV payload write
- NO → Pay KV write; append live cold row

c) Cold overflow: reclaim shadows, then LRU live rows
Drop permanently; preserve the live floor where feasible

Emit Next Token

Shadows: reclaim the backup whose hot twin has the highest cumulative score.
Cold storage: GPU by default; pinned CPU option. STT costs are analytical.
Standalone sketch/peek simulation is a separate path; it is not used in this loop.